PROJECT REPORT

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🔹 1. Problem Statement

Managing daily expenses is an important task, but many people do not keep proper records of their spending. This can lead to poor financial planning and unnecessary expenses. The problem is to create a simple system that helps users track their expenses and calculate their total spending easily.

🔹 2. Importance of the Problem

Expense tracking helps individuals understand where their money is being spent. It improves financial awareness and helps in better budgeting. For students and beginners, it is also a practical way to apply programming concepts to real-life problems.

🔹 3. Objective

- To develop a simple expense tracking system
- To calculate monthly and yearly expenses
- To apply basic Python concepts in a real-world scenario

🔹 4. Approach / Methodology

The project is implemented using a **menu-driven program** in Python. A list is used to store expense data temporarily. Each expense contains three values: month, name, and amount.

The program includes three main functions:

- **Add Expense:** Stores expense details in the list
- **Monthly Total:** Calculates total expenses for a selected month
- **Yearly Total:** Calculates overall expenses

The user interacts with the program by selecting options from a menu.

🔹 5. Key Design Decisions

- Used **list** instead of file/database for simplicity

- Implemented **functions** to organize code
- Used **loops and conditions** to process data
- Designed a **menu interface** for easy interaction

🔹 6. Challenges Faced

- Handling empty data (when no expenses are added)
- Ensuring correct matching of months (case sensitivity)
- Understanding how to store and retrieve data using lists

🔹 7. Solution

The problem is solved by allowing users to input expenses and storing them in a list. The program then processes this data using loops to calculate monthly and yearly totals. This provides a simple and effective way to track expenses.

🔹 8. Limitations

- Data is not stored permanently (lost after program ends)
- No graphical user interface
- Limited features

🔹 9. Learning Outcomes

- Understanding of Python basics (lists, loops, functions)
- Experience in solving real-life problems using code
- Improved logical thinking and problem-solving skills

🔹 10. Conclusion

The Expense Tracker project successfully demonstrates how Python can be used to solve a real-world problem. It is a simple and effective tool for tracking expenses and understanding spending patterns. The project can be further improved by adding more features and permanent data storage.